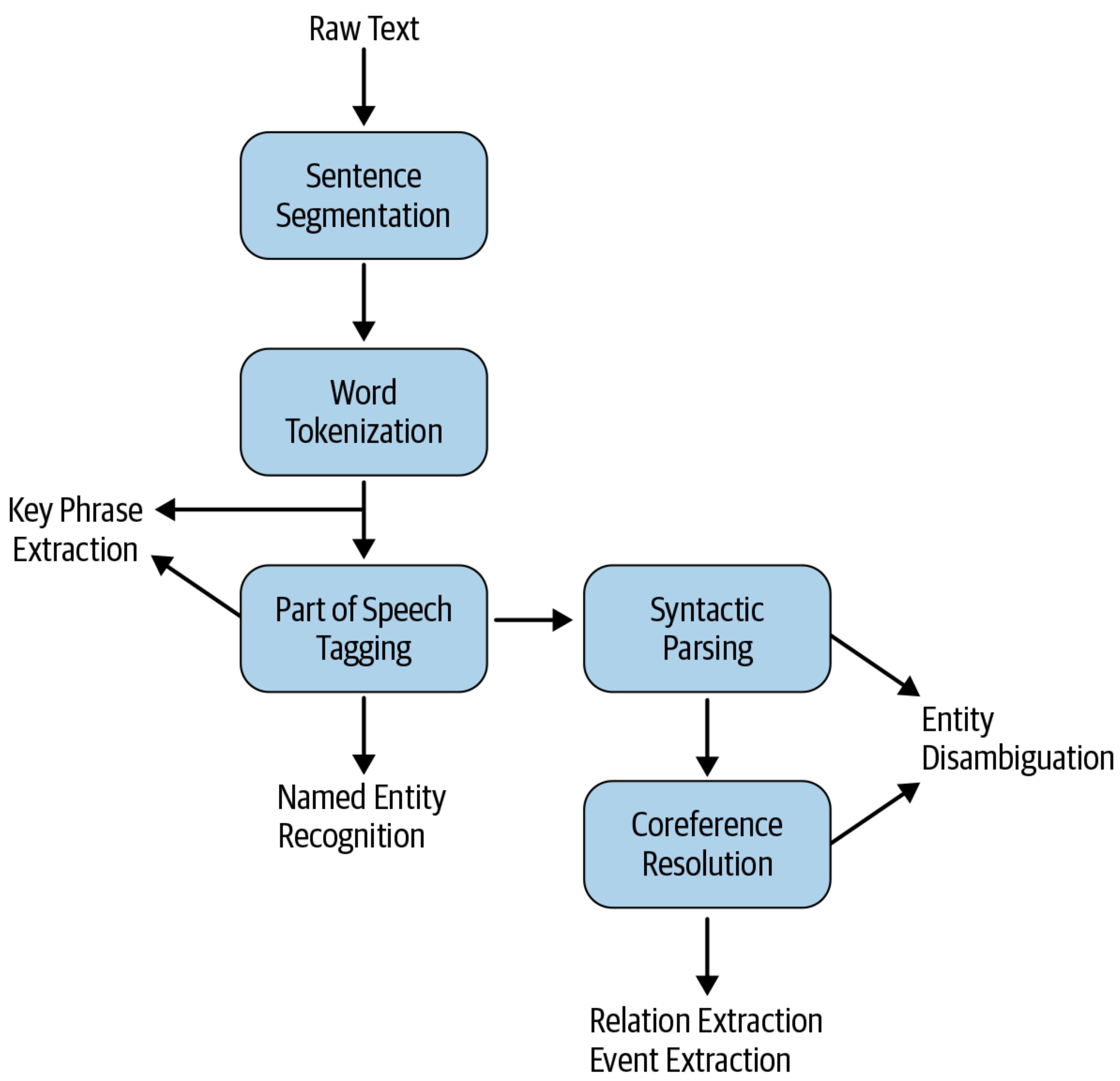
Applied NLP

DATA 641

Lecture 10 — Information Extraction; General pipeline

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

The general pipeline for IE



The general pipeline for IE requires more fine-grained NLP processing than what we saw for text classification. However, not all steps in the pipeline are necessary for all IE tasks, and the figure demonstrates which IE tasks require what levels of analysis.

- To identify named entities (persons, organizations, etc.), we would need to know the part-of-speech tags of words
- For relating multiple references to the same entity (e.g., Albert Einstein, Einstein, the scientist, he, etc.), we would need coreference resolution.
- Key phrase extraction is the task requiring minimal NLP processing (some algorithms also do POS tagging before extracting key-phrases)
- IE tasks are typically evaluated in terms of precision, recall, and F1 scores using standard evaluation sets
- Considering the different levels of NLP pre-processing required, IE tasks are also affected by the accuracy of these processing steps themselves.
- Collecting relevant training data and training our own models for IE, if necessary, should take all these aspects into account.